

Appendix A

Vector algebra and calculus

Vector products

$$\left. \begin{aligned} \mathbf{A} \cdot \mathbf{B} &= A_x B_x + A_y B_y + A_z B_z \\ \mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \end{aligned} \right\} \text{ in Cartesian coordinates}$$

Vector triple products

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} \quad \text{in Cartesian coordinates} \quad (\text{A.1})$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad (\text{A.2})$$

Vector derivatives

The vector derivative operator is denoted by ∇ (pronounced “nabla”). It can mean (i) the *gradient*, ∇ (when acting on a scalar function or “field”); (ii) the *divergence*, $\nabla \cdot$ (when dotted with a vector field following it); or (iii) the *curl*, $\nabla \times$ (when crossed with a vector field following it).

In Cartesian coordinates, for all of the three cases, ∇ can be written out as:

$$\nabla \equiv \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}$$

Vector derivatives of products

$$\nabla(fg) = f(\nabla g) + g(\nabla f) \quad (\text{A.3})$$

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} \quad (\text{A.4})$$

$$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f) \quad (\text{A.5})$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \quad (\text{A.6})$$

$$\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f) \quad (\text{A.7})$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) \quad (\text{A.8})$$

Second vector derivatives

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0 \quad (\text{A.9})$$

$$\nabla \times (\nabla f) = \mathbf{0} \quad (\text{A.10})$$

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (\text{A.11})$$

Fundamental theorems¹ for vector derivatives

$$\int_a^b (\nabla f) \cdot d\mathbf{l} = f(\mathbf{b}) - f(\mathbf{a}) \quad \text{fundamental theorem for gradients} \quad (\text{A.12})$$

$$\int_V (\nabla \cdot \mathbf{A}) d\tau = \oint_{S \equiv \partial V} \mathbf{A} \cdot d\mathbf{a} \quad \text{Gauss' divergence theorem, Green's theorem} \quad (\text{A.13})$$

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint_{P \equiv \partial S} \mathbf{A} \cdot d\mathbf{l} \quad \text{Stokes' theorem} \quad (\text{A.14})$$

¹Recall the fundamental theorem of calculus: $\int_a^b f'(x) dx = f(b) - f(a)$.

Useful expressions for radial vector derivatives

$$\nabla (r^n) = n r^{n-1} \hat{\mathbf{r}} \quad (\text{A.15})$$

$$\nabla \cdot (r^n \hat{\mathbf{r}}) = (n+2) r^{n-1} \quad \text{for } n \neq -2$$

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta^{(3)}(\mathbf{r}) \quad (\text{A.16})$$

$$\nabla \times (r^n \hat{\mathbf{r}}) = \mathbf{0}. \quad (\text{A.17})$$

Index notation / tensor notation

Vectors: $\mathbf{A} \rightarrow A_i, \mathbf{B} \rightarrow B_j$, etc.

Dot products: $\mathbf{A} \cdot \mathbf{B} \rightarrow \sum_{ij} \delta_{ij} A_i B_j = \sum_i A_i B_i \equiv A_i B_i$ (Einstein summation convention) (A.18)

Cross products: $\mathbf{A} \times \mathbf{B} \rightarrow \sum_{jk} \varepsilon_{ijk} A_j B_k \equiv \varepsilon_{ijk} A_j B_k$ (A.19)

The “Levi-Civita symbol” ε_{ijk} is the fully antisymmetric quantity with $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = 1$, and $\varepsilon_{321} = \varepsilon_{213} = \varepsilon_{132} = -1$. It satisfies the identity:

$$\sum_i \varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}. \quad (\text{A.20})$$

This identity leads to the expression Eq. (A.2) for the vector triple product. The index notation can be extended to gradients, divergences, and curls with the substitution $\nabla \rightarrow \partial_i$. All the identities involving vector derivatives Eqs. (A.3–A.11) follow from there in a straightforward manner.

Appendix B

The Dirac δ function

The Dirac δ function is defined as,

$$\delta(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ \infty & \text{if } x = 0, \end{cases}$$

such that,

$$\int_{-\infty}^{\infty} \delta(x) dx = 1. \quad (\text{B.1})$$

The three dimensional Dirac δ function is understood as,

$$\delta^3(\mathbf{r}) \equiv \delta(x) \delta(y) \delta(z). \quad (\text{B.2})$$

Of prime consequence in electrodynamics, is the following expression:

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta^3(\mathbf{r}). \quad (\text{B.3})$$

Appendix C

Expressions for vector derivatives

Cartesian. $dl = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}; \quad d\tau = dx dy dz.$

$$\nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{x}} + \frac{\partial f}{\partial y} \hat{\mathbf{y}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}} \quad (\text{C.1})$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad (\text{C.2})$$

$$\nabla \times \mathbf{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{\mathbf{z}} \quad (\text{C.3})$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \quad (\text{C.4})$$

Cylindrical. $dl = d\rho \hat{\boldsymbol{\rho}} + \rho d\phi \hat{\boldsymbol{\phi}} + dz \hat{\mathbf{z}}; \quad d\tau = \rho d\rho d\phi dz.$

$$\nabla f = \frac{\partial f}{\partial \rho} \hat{\boldsymbol{\rho}} + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \hat{\boldsymbol{\phi}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}} \quad (\text{C.5})$$

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z} \quad (\text{C.6})$$

$$\nabla \times \mathbf{A} = \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \hat{\boldsymbol{\rho}} + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right] \hat{\boldsymbol{\phi}} + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (\rho A_\phi) - \frac{\partial A_\rho}{\partial \phi} \right] \hat{\mathbf{z}} \quad (\text{C.7})$$

$$\nabla^2 f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2} \quad (\text{C.8})$$

Spherical. $dl = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}}; \quad d\tau = r^2 \sin \theta dr d\theta d\phi.$

$$\nabla f = \frac{\partial f}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\boldsymbol{\phi}} \quad (\text{C.9})$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi} \quad (\text{C.10})$$

$$\begin{aligned} \nabla \times \mathbf{A} = & \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right] \hat{\mathbf{r}} \\ & + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \hat{\boldsymbol{\theta}} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\boldsymbol{\phi}} \end{aligned} \quad (\text{C.11})$$

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \quad (\text{C.12})$$

Appendix D

Maxwell's equations

Maxwell's equations in differential form¹

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho \quad (\text{Gauss' law}) \quad (\text{D.1})$$

$$\nabla \cdot \mathbf{B} = 0 \quad (\text{absence of magnetic monopole}) \quad (\text{D.2})$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (\text{Faraday's law}) \quad (\text{D.3})$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (\text{Ampère-Maxwell law}) \quad (\text{D.4})$$

Lorentz force law

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (\text{D.5})$$

Fundamental constants in relativity and electromagnetism

$$\text{Speed of light, } c \equiv 2.99792458 \times 10^8 \text{ ms}^{-1} \quad (\text{D.6})$$

$$\text{Permeability of free space, } \mu_0 \equiv 4\pi \times 10^{-7} \text{ NA}^{-2} \quad (\text{D.7})$$

$$\text{Permittivity of free space, } \epsilon_0 \equiv \frac{1}{\mu_0 c^2} \quad (\text{D.8})$$

$$\frac{1}{4\pi\epsilon_0} \approx 8.89 \times 10^9 \text{ Nm}^2\text{C}^{-2} \quad (\text{D.9})$$

Maxwell's equations in integral form

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enclosed}} \quad (\text{Gauss' law}) \quad (\text{D.10})$$

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0 \quad (\text{absence of magnetic monopole}) \quad (\text{D.11})$$

$$\oint_{\partial S} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a} \quad (\text{Faraday's law}) \quad (\text{D.12})$$

$$\oint_{\partial S} \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}} + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{a} \quad (\text{Ampère-Maxwell law}) \quad (\text{D.13})$$

¹We use SI units throughout this course.

Appendix E

Four-vectors and four-tensors

A contravariant vector v^μ transforms as $v'^\mu = \Lambda^\mu_\alpha v^\alpha$.¹ A covariant vector a_μ transforms as $a'_\mu = \bar{\Lambda}^\alpha_\mu a_\alpha$. Here Λ^μ_ν is the Lorentz transformation matrix and $\bar{\Lambda}^\mu_\nu$ its inverse with the property $\bar{\Lambda}^\mu_\nu \Lambda^\nu_\sigma = \Lambda^\mu_\nu \bar{\Lambda}^\nu_\sigma = \delta^\mu_\sigma$.

For boosts in the x -direction, the components of these matrices are:

$$\Lambda^\mu_\nu \equiv \frac{\partial x'^\mu}{\partial x^\nu} = \begin{pmatrix} \gamma & -\gamma\beta & & \\ -\gamma\beta & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix}, \quad \bar{\Lambda}^\mu_\nu \equiv \frac{\partial x^\mu}{\partial x'^\nu} = \begin{pmatrix} \gamma & \gamma\beta & & \\ \gamma\beta & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix}. \quad (\text{E.1})$$

More explicitly,

$$\begin{array}{llll} v'^0 = \gamma(v^0 - \beta v^1) & v^0 = \gamma(v'^0 + \beta v'^1) & a'_0 = \gamma(a_0 + \beta a_1) & a_0 = \gamma(a'_0 - \beta a'_1) \\ v'^1 = \gamma(v^1 - \beta v^0) & v^1 = \gamma(v'^1 + \beta v'^0) & a'_1 = \gamma(a_1 + \beta a_0) & a_1 = \gamma(a'_1 - \beta a'_0) \\ v'^2 = v^2 & v^2 = v'^2 & a'_2 = a_2 & a_2 = a'_2 \\ v'^3 = v^3 & v^3 = v'^3 & a'_3 = a_3 & a_3 = a'_3. \end{array} \quad (\text{E.2})$$

It follows that the dot product of a covariant and a contravariant vector such as $a_\alpha v^\alpha$ is a *scalar* or an *invariant* under (Lorentz) transformations.

A tensor is similarly defined by its associated transformation property. A tensor of rank (p, q) has p contravariant and q covariant indices and transforms as

$$T'^{\mu\nu\dots}_{\kappa\lambda\dots} = \Lambda^\mu_\alpha \Lambda^\nu_\beta \dots \bar{\Lambda}^\gamma_\kappa \bar{\Lambda}^\delta_\lambda \dots T^{\alpha\beta\dots}_{\gamma\delta\dots}, \quad (\text{E.3})$$

where p factors of Λ and q factors of $\bar{\Lambda}$ appear.

The metric: lowering and raising indices

The metric $\eta_{\mu\nu}$ is a tensor of rank $(0, 2)$, which defines a dot product between two contravariant vectors, and helps measure the “invariant distance” on a four-dimensional spacetime.

$$ds^2 = -(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2 \equiv \eta_{\mu\nu} dx^\mu dx^\nu. \quad (\text{E.4})$$

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}, \quad \text{and} \quad \eta^{\mu\nu} = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \quad \text{with} \quad \eta^{\mu\nu} \eta_{\nu\sigma} = \delta^\mu_\sigma. \quad (\text{E.5})$$

The metric $\eta_{\mu\nu}$ and its inverse $\eta^{\mu\nu}$ can also be used to respectively lower or raise indices, and thus convert a contravariant index into a covariant one, or vice-versa.

$$v_\mu \equiv \eta_{\mu\nu} v^\nu, \quad a^\mu \equiv \eta^{\mu\nu} a_\nu. \quad (\text{E.6})$$

¹We employ the Einstein summation convention: repeated indices are summed over.

Appendix F

Relativistic mechanics and electrodynamics

Kinematics and dynamics

For massive objects: proper velocity, $\eta^\mu \equiv \frac{dx^\mu}{d\tau} = (\gamma c, \gamma \mathbf{v}) = (\gamma c, \gamma v_x, \gamma v_y, \gamma v_z)$; proper momentum, $p^\mu \equiv m\eta^\mu = (\gamma mc, \gamma m\mathbf{v})$. For light: $E = pc$ and $p^\mu = (E/c, 0, 0, E/c)$ if propagating in the z -direction. In general: $E^2 = p^2 c^2 + m^2 c^4$.

Newton's second law reads (with p^i as components of proper momentum),

$$F^i = \frac{dp^i}{dt} \quad \Rightarrow \quad K^\mu = \frac{dp^\mu}{d\tau}, \quad (\text{F.1})$$

where $K^\mu \equiv (\frac{1}{c} \frac{dE}{d\tau}, \gamma \mathbf{F})$ is the *Minkowski force*.

Electrodynamics

The electric and magnetic fields are contained in the *field tensor* $F^{\mu\nu}$ defined as

$$F^{\mu\nu} \equiv \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}. \quad (\text{F.2})$$

Note that this can also be written as

$$F^{0i} = \frac{1}{c} E_i, \quad F^{ij} = \varepsilon_{ijk} B_k. \quad (\text{F.3})$$

One can write a four-dimensional generalization of the Levi-Civita symbol with $\varepsilon^{0123} = 1$, and other components defined by (anti)symmetry. One can then also define the *dual* tensor,

$$\tilde{F}^{\mu\nu} \equiv \frac{1}{2} \varepsilon^{\mu\nu\kappa\lambda} F_{\kappa\lambda} = \begin{pmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & -E_z/c & E_y/c \\ -B_y & E_z/c & 0 & -E_x/c \\ -B_z & -E_y/c & E_x/c & 0 \end{pmatrix}. \quad (\text{F.4})$$

Finally, once can define a **four-current density**,

$$J^\mu \equiv (c\rho, \mathbf{J}) = (c\rho, J_x, J_y, J_z), \quad \text{which satisfies the } \mathbf{continuity equation}, \quad \partial_\mu J^\mu = 0. \quad (\text{F.5})$$

The **Maxwell's equations** read,

$$\partial_\nu F^{\mu\nu} = \mu_0 J^\mu, \quad \partial_\nu \tilde{F}^{\mu\nu} = 0. \quad (\text{F.6})$$

Generalization of the **Lorentz force law**. For a particle of charge q , moving with a proper velocity η^μ , the Minkowski force K^μ on it is given by

$$K^\mu = q\eta_\nu F^{\mu\nu}. \quad (\text{F.7})$$

Potential formulation

Due to absence of magnetic charge, a four-potential A^μ can defined as $A^\mu \equiv (V/c, \mathbf{A}) = (V/c, A_x, A_y, A_z)$, which gives the field tensor via $F^{\mu\nu} \equiv \partial^\mu A^\nu - \partial^\nu A^\mu$. The Lorenz gauge condition becomes $\partial_\mu A^\mu = 0$.

Appendix G

Legendre polynomials, multipole expansion

Legendre polynomials

$$\begin{aligned}P_0(x) &= 1 \\P_1(x) &= x \\P_2(x) &= \frac{1}{2}(3x^2 - 1) \\P_3(x) &= \frac{1}{2}(5x^3 - 3x) \\P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3) \\P_5(x) &= \frac{1}{8}(63x^5 - 70x^3 + 15x).\end{aligned}\tag{G.1}$$

Orthogonality

$$\int_{-1}^1 P_l(x) P_{l'}(x) dx = \int_0^\pi P_l(\cos \theta) P_{l'}(\cos \theta) \sin \theta d\theta = \frac{2}{2l+1} \delta_{ll'}.\tag{G.2}$$

Separation of variables in spherical coordinates

With azimuthal symmetry, a general solution to the Laplace equation can be written as

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta).\tag{G.3}$$

Multipole expansion

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}') d\tau'}{r} = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{r^{l+1}} \int (r')^l P_l(\cos \theta') \rho(\mathbf{r}') d\tau'.\tag{G.4}$$

$$\text{The monopole moment,} \quad Q \equiv \int \rho(\mathbf{r}') d\tau',\tag{G.5}$$

$$\text{The dipole moment,} \quad \mathbf{p} \equiv \int \mathbf{r}' \rho(\mathbf{r}') d\tau',\tag{G.6}$$

$$\text{The quadrupole moment,} \quad Q_{ij} \equiv \int \frac{1}{2} [3r'_i r'_j - (r')^2 \delta_{ij}] \rho(\mathbf{r}') d\tau'.\tag{G.7}$$

Appendix H

Maxwell's equations in matter

$$\nabla \cdot \mathbf{D} = \rho_f, \quad (\text{H.1})$$

$$\nabla \cdot \mathbf{B} = 0, \quad (\text{H.2})$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (\text{H.3})$$

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}. \quad (\text{H.4})$$

Constitutive relations

$$\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P}, \quad (\text{H.5})$$

$$\mathbf{H} \equiv \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}. \quad (\text{H.6})$$

Linear media

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad \text{with} \quad \epsilon \equiv \epsilon_0 (1 + \chi_e), \quad \mathbf{D} = \epsilon \mathbf{E}, \quad (\text{H.7})$$

$$\mathbf{M} = \chi_m \mathbf{H}, \quad \text{with} \quad \mu \equiv \mu_0 (1 + \chi_m), \quad \mathbf{H} = \frac{1}{\mu} \mathbf{B}. \quad (\text{H.8})$$

Bound charge and current densities

$$\rho_b = -\nabla \cdot \mathbf{P} \quad \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}, \quad (\text{H.9})$$

$$\mathbf{J}_b = \nabla \times \mathbf{M} \quad \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}. \quad (\text{H.10})$$

Boundary conditions

$$D_1^\perp - D_2^\perp = \sigma_f \quad B_1^\perp - B_2^\perp = 0 \quad (\text{H.11})$$

$$\mathbf{E}_1^\parallel - \mathbf{E}_2^\parallel = \mathbf{0} \quad \mathbf{H}_1^\parallel - \mathbf{H}_2^\parallel = \mathbf{K}_f \times \hat{\mathbf{n}}. \quad (\text{H.12})$$